MILITARY STANDARD

MIL-STD-1320-12A
(NAVY)
21 MAY 1991

TRUCKLOADING

TORPEDO MK 44 AND MK 46 ASSEMBLIES

IN CONTAINER MK 197 MOD 1

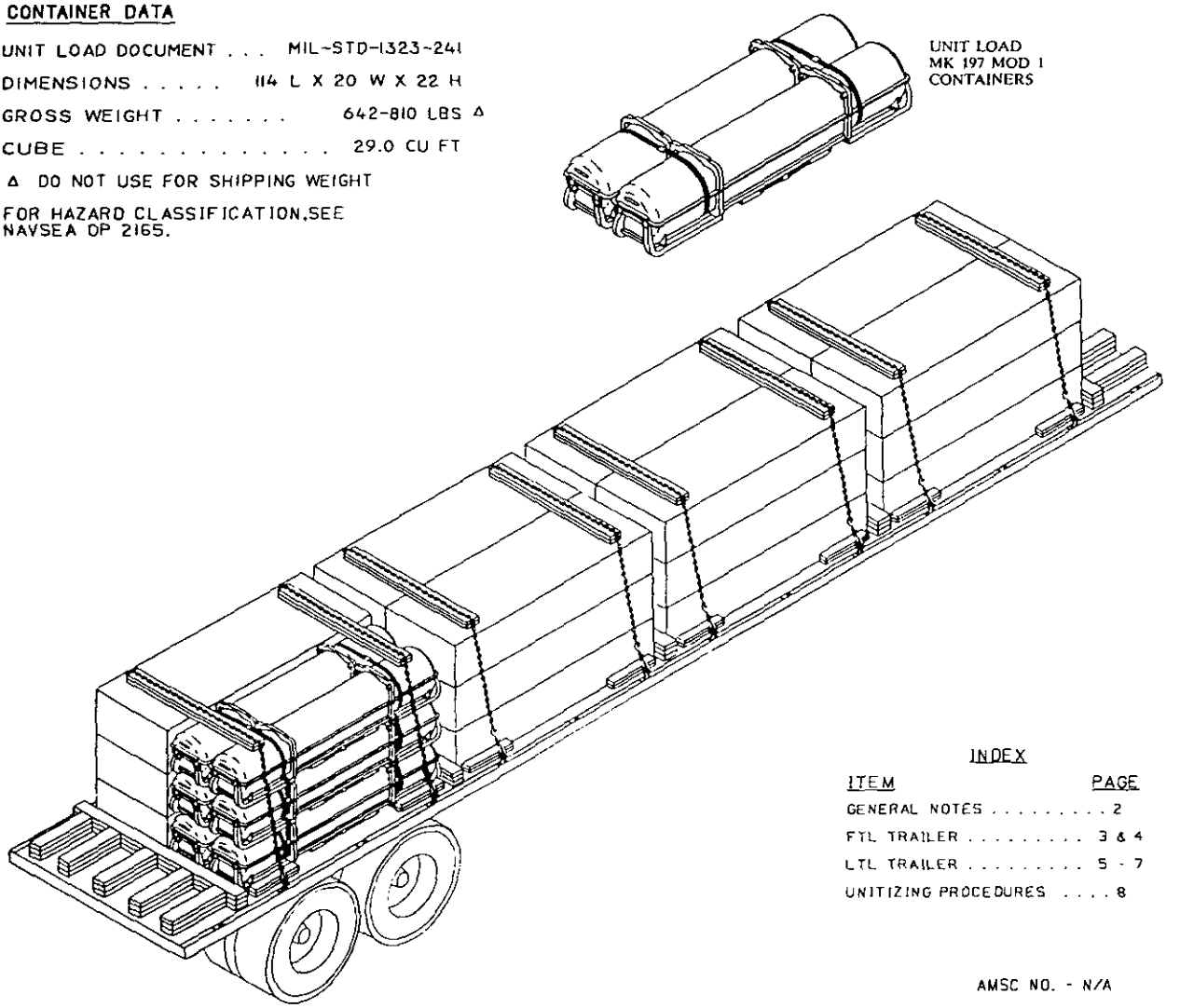
SUPERSEDING
MIL-STD-1320-12
3 JANUARY 1975

CONTAINER DATA

UNIT LOAD DOCUMENT . . . MIL-STD-1323-241
DIMENSIONS . . . . . 114 L X 20 W X 22 H
GROSS WEIGHT . . . . . 642-810 LBS Δ
CUBE . . . . . 29.0 CU FT

Δ DO NOT USE FOR SHIPPING WEIGHT

FOR HAZARD CLASSIFICATION, SEE NAVSEA OP 2165.



UNIT LOAD
MK 197 MOD 1
CONTAINERS

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AMSC NO. - N/A

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NOTES:

1. UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS. MIL-HDBK-236 (NAVY).

REV LTR	REVISION DESCRIPTION	DATE	TOA	SYSCOM
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FSC 8140

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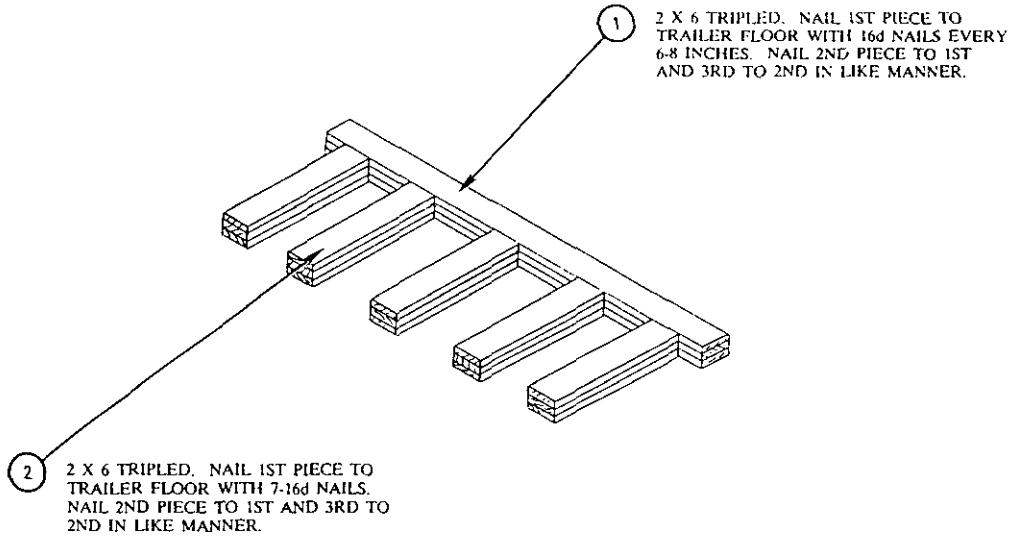
ORIGINATOR — DATE 21 MAY 91

WPNSTA EARLE. NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING LOADED CONTAINERS AND UNIT LOADS OF MK 44 AND MK 46 TORPEDO ASSEMBLIES IN THE MK 197 MOD 1 CONTAINER.
2. THE PROCEDURES DESCRIBED HEREIN ARE INTENDED FOR 45 FT THROUGH 48 FT FLATBED TRAILERS WITH THE TRAILERS TANDEM AXLES LOCATED IN ANY POSITION. DO NOT USE TRAILERS WITH METAL FLOORS.
3. CHAINS AND STEEL STRAPPING MAY BE USED INTERCHANGEABLY ON A ONE TO ONE BASIS FOR TIE DOWNS.
4. STEEL STRAPPING USED AS TIE DOWNS SHALL BE 2 X .050 AND SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).
5. CHAINS, FITTINGS AND LOAD BINDERS SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY) EXCEPT THAT THE CHAIN/GRAB HOOKS SHALL BE ATTACHED TO THE STAKE POCKETS (NOT AROUND THE RUB RAIL).
6. PROTECTOR BOARDS (DOUBLED 2 X 6'S) SHALL BE USED UNDER THE TIE DOWNS. WHEN USING STEEL STRAPPING, THE STRAPS SHALL BE STAPLED TO THE PROTECTOR BOARD USING 2 INCH STRAPPING STAPLES. WHEN USING CHAIN, THE CHAINS SHALL BE SECURED TO THE PROTECTOR BOARD BY DRIVING A 10d NAIL THROUGH THE CHAIN LINK AND CLINCHING THE NAIL OVER THE CHAIN. FIVE STAPLES/NAILS SHALL BE USED FOR EACH TIE DOWN.
7. PRIOR TO LOADING THE TRAILER AND DURING PRELOADING INSPECTION, REQUIRED BY NAVSEA OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS AND LOAD BINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS, FITTINGS AND LOAD BINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOAD BINDER.
8. UNLESS OTHERWISE SPECIFIED, DUNNAGE LUMBER, STEEL STRAPPING, STRAP SEALS AND NAILS SHALL BE AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).
9. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
10. AFTER THE BLOCKING, BRACING AND TIE DOWNS HAVE BEEN INSPECTED, THE CONTAINERS SHALL BE COMPLETELY COVERED WITH FIRE RESISTANT WATERPROOF TARPAULINS. THE TARPAULINS MAY BE UNDER THE TIE DOWNS.
11. AFTER THE TARPAULINS ARE IN PLACE, ATTACH PLACARDS AS REQUIRED TO BOTH SIDES, FRONT, AND BACK OF THE TRAILER AND ATTACH THE SHIPPING DOCUMENTS TO AN ACCESSABLE AREA ON THE BACK DECK OF THE TRAILER.
12. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).



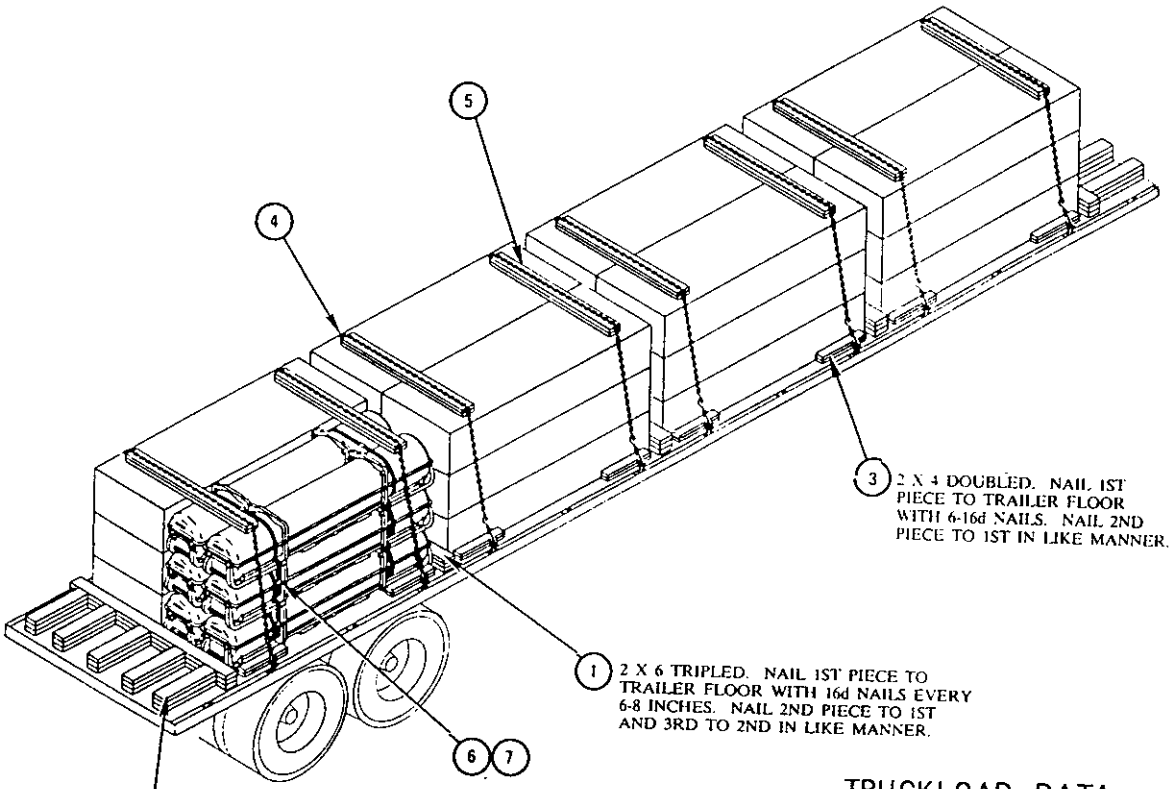
DETAIL A

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FTL 45 FT THROUGH 48 FT TRAILER (FLATBED)
(45 FT TRAILER SHOWN)

NOTE

TARPAULINS OVER CONTAINERS OMITTED FOR CLARITY.



TRUCKLOAD DATA

NUMBER OF CONTAINERS 48

WEIGHT . . . . . 30816-38880 LBS Δ

Δ DO NOT USE FOR SHIPPING WEIGHT

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
32	7	SEAL	STEEL	1 1/4 X .035
32	6	STRAP	STEEL	1 1/4 X .035 X 11 FT
8	5	CHAIN AND LOAD BINDER	SEE NOTE 5 ON PAGE 2	5/16 OR 3/8
16	4	PROTECTOR BOARD	WOOD	2 X 6 X 70
32	3	SLEEPER	WOOD	2 X 4 X 24
30	2	BACKUP CLEAT	WOOD	2 X 6 X 18
15	1	CROSSMEMBER	WOOD	2 X 6 X 84

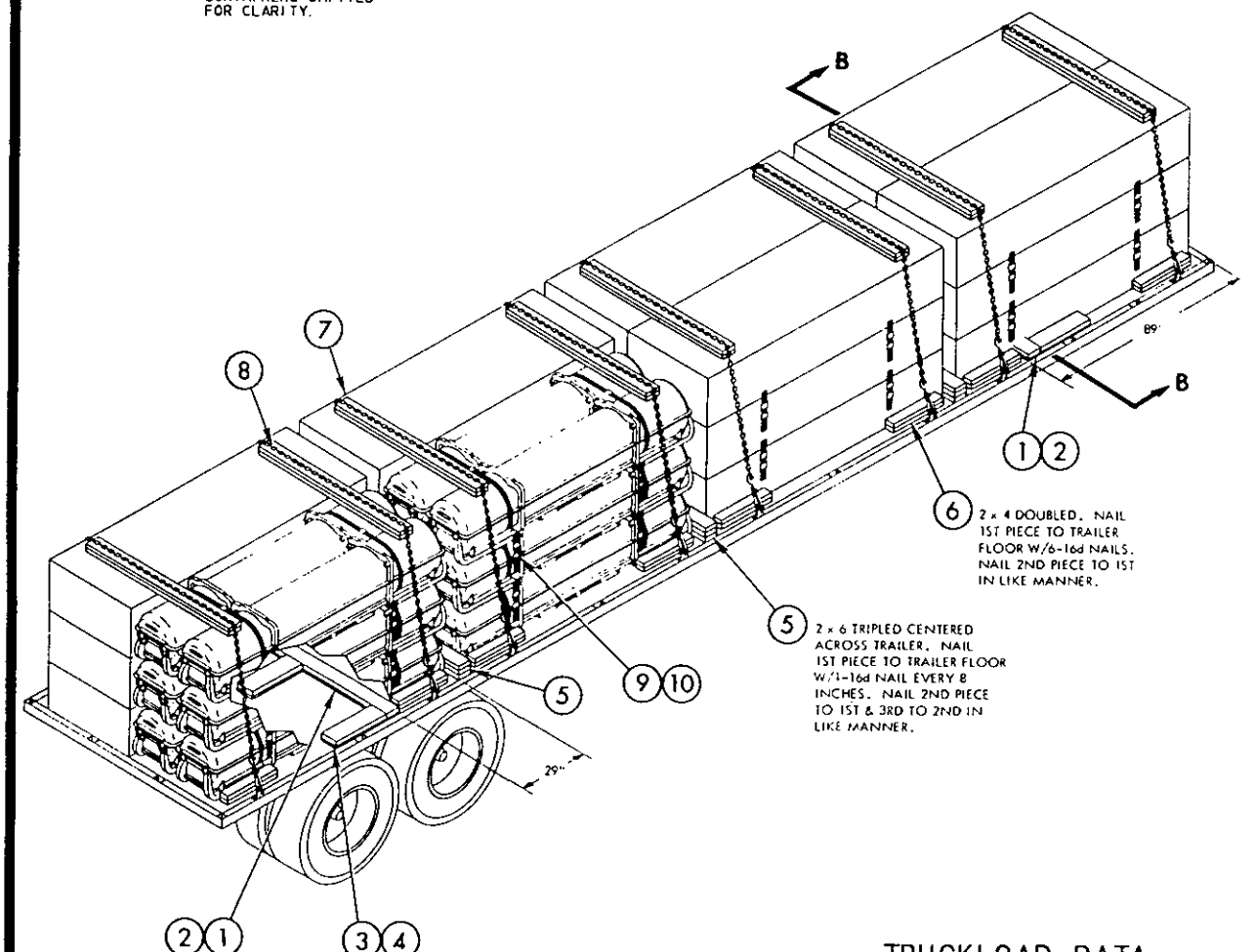
LIST OF MATERIALS

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FTL 40 FT THROUGH 44 FT TRAILER (FLATBED)
(40 FT TRAILER SHOWN)

NOTE

TARPAULINS OVER CONTAINERS OMITTED FOR CLARITY.



6 2 x 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

5 2 x 6 TRIPLED CENTERED ACROSS TRAILER. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST & 3RD TO 2ND IN LIKE MANNER.

2 1 NAIL PIECE 1 TO TRAILER FLOOR W/1-10d NAIL EVERY 8 INCHES. NAIL PIECE 2 TO PIECE 1 W/1-16d NAIL EVERY 8 INCHES.

3 4 NAIL PIECE 3 TO TRAILER FLOOR W/1-10d NAIL EVERY 6 INCHES, NAIL PIECE 4 TO PIECE 3 W/1-16d NAIL EVERY 6 INCHES.

TRUCKLOAD DATA

NUMBER OF CONTAINERS 48

WEIGHT . . . . . 30816-38880 LBS Δ

Δ DO NOT USE FOR SHIPPING WEIGHT

LIST OF MATERIALS

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
32	10	SEAL	STEEL	5/16 OR 3/8
32	9	STRAP	STEEL	5/16 OR 3/8
8	8	CHAIN AND LOAD BINDER	STEEL	5/16 OR 3/8
16	7	PROTECTOR BOARD	WOOD	2 X 6 X 70
32	6	SLEEPER	WOOD	2 X 4 X 24
9	5	CROSSMEMBER	WOOD	2 X 6 X 90
6	4	TOP CLEAT	WOOD	2 X 8 X 24
6	3	BOTTOM CLEAT	WOOD	1 X 8 X 24
2	2	TOP RESTRAINER	WOOD	2 X 6 X 90
2	1	BOTTOM RESTRAINER	WOOD	1 X 6 X 90

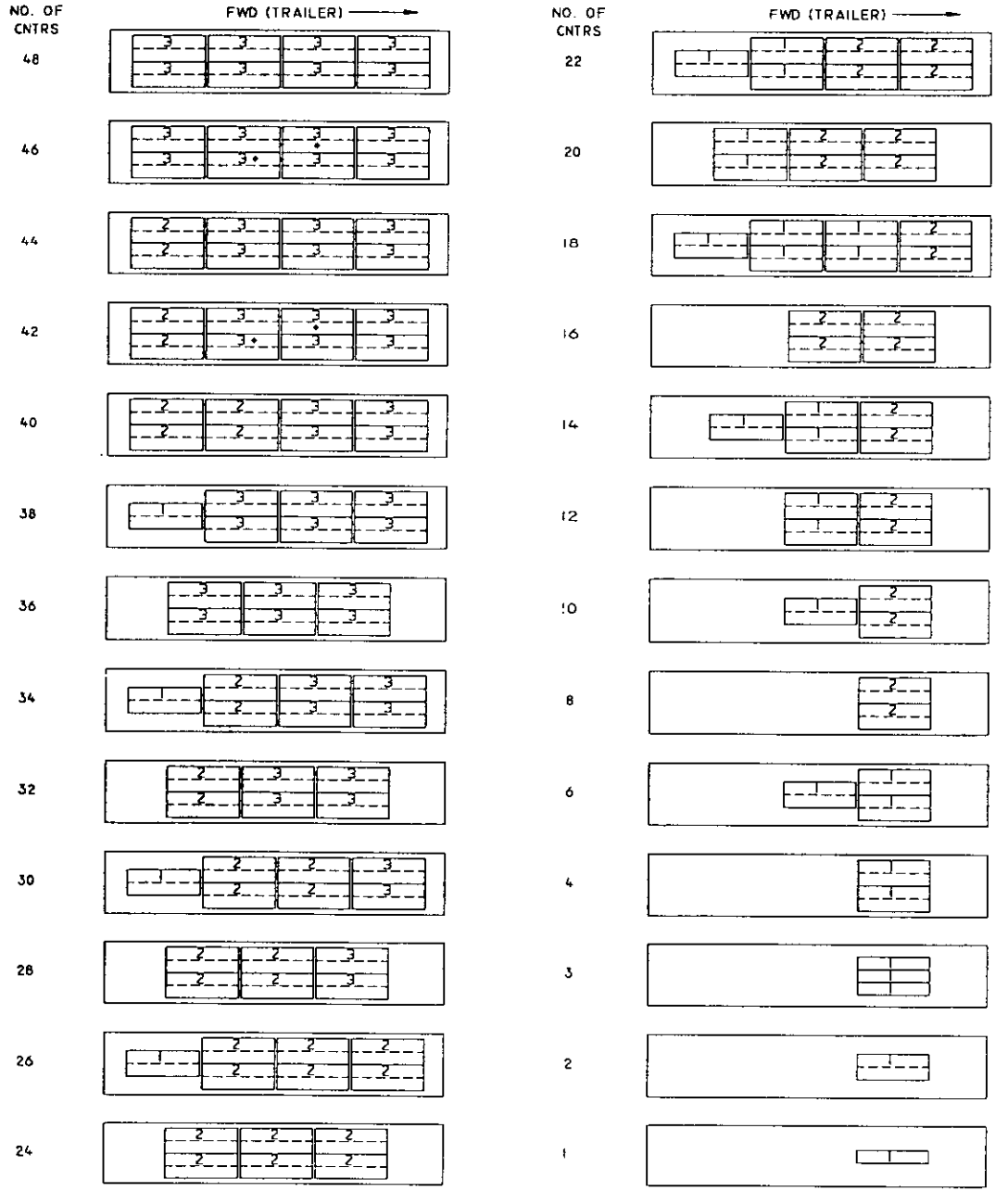
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LTL 40 FT THROUGH 48 FT TRAILERS (FLATBED)

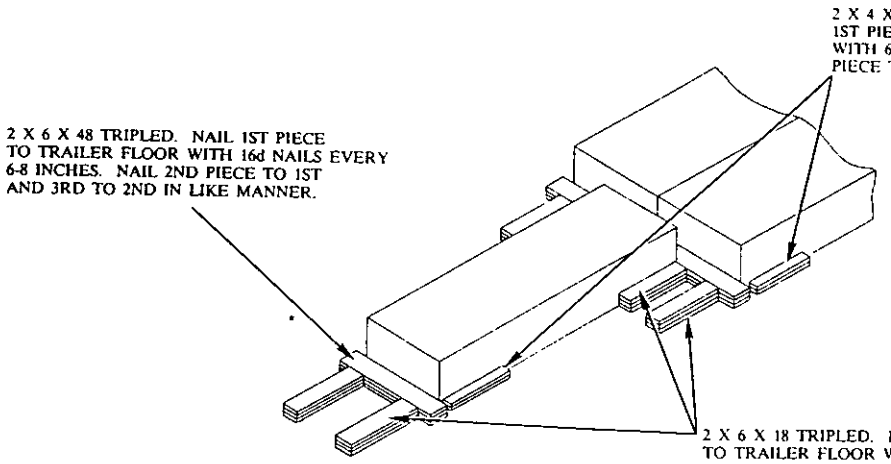
1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD, SELECT THE CORRECT LOAD PATTERN, FOR THE NUMBER OF UNIT LOADS TO BE SHIPPED, FROM TABLE I.
2. POSITION UNIT LOADS ON THE TRAILER AS SHOWN IN TABLE I. BLOCK AND TIE DOWN USING THE PRINCIPLES SHOWN ON PAGES 6 AND 7 AND THE GENERAL DOCUMENT MIL-STD-1320 (NAVY).
3. THE BLOCKING ON PAGE 6 SHOWS THE INTERFACE BETWEEN STACKS OF ONE AND TWO UNIT LOADS.
4. WHEN LOADING AN ODD NUMBER OF CONTAINERS OR FOR SHIPPING 42 OR 46 CONTAINERS, REPLACE A TOP INSIDE CONTAINER WITH THE FILLER ASSEMBLY SHOWN ON PAGE 7 AND INDICATED BY A ASTERISK BELOW.

TABLE I

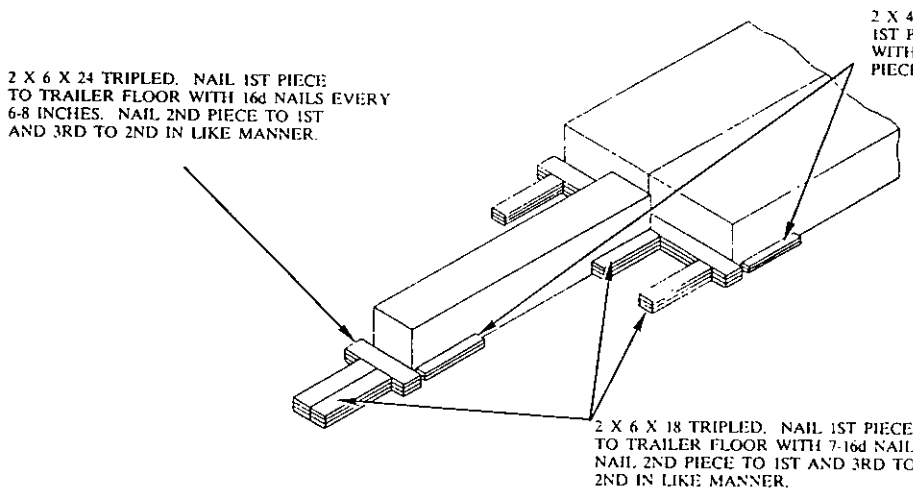
LOAD PATTERN *



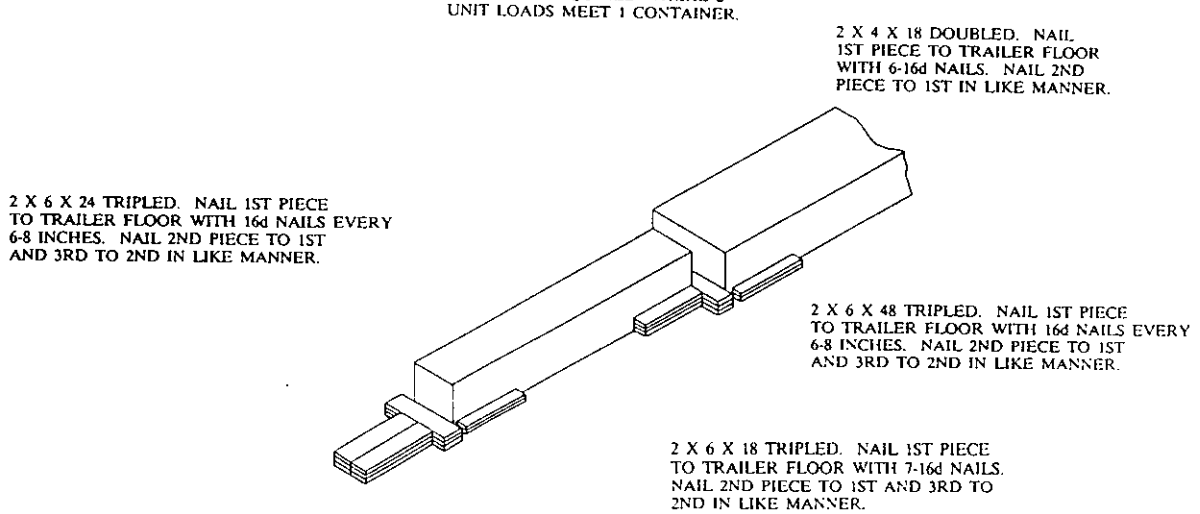
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BLOCKING REQUIRED WHERE 2 UNIT LOADS MEET 1 UNIT LOAD.



BLOCKING REQUIRED WHERE 2 UNIT LOADS MEET 1 CONTAINER.

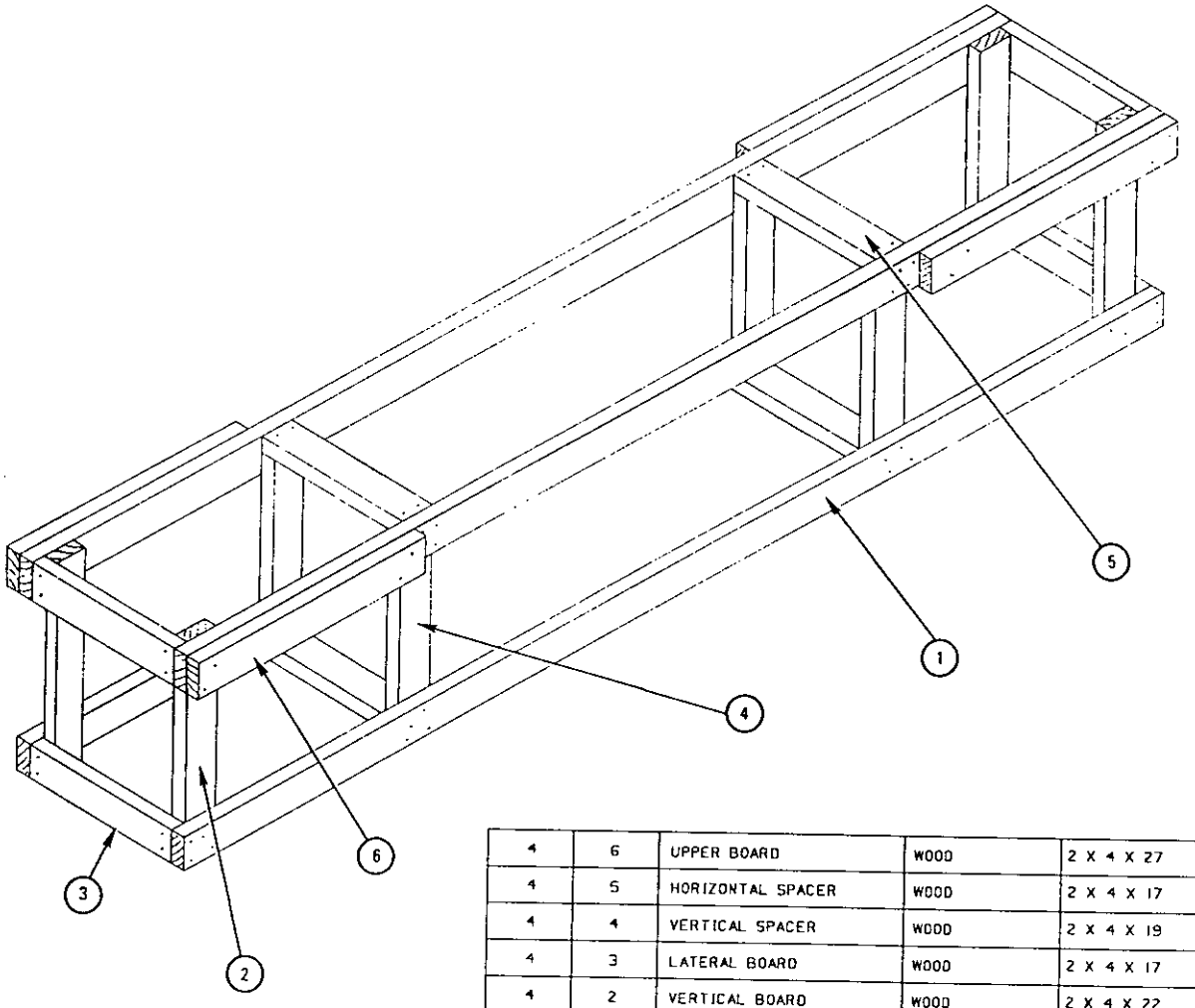


BLOCKING REQUIRED WHERE 1 UNIT LOAD MEETS 1 CONTAINER.

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FILLER ASSEMBLY INSTRUCTIONS

1. END NAIL LONGITUDINAL BOARD, ITEM 1, TO LATERAL BOARD, ITEM 3, WITH 2-10d NAILS.
2. END NAIL LONGITUDINAL BOARD, ITEM 1, TO HORIZONTAL SPACER, ITEM 5, WITH 2-10d NAILS.
3. END NAIL HORIZONTAL SPACER, ITEM 5, TO VERTICAL SPACER, ITEM 4, WITH 2-10d NAILS.
4. NAIL LONGITUDINAL BOARD, ITEM 1, TO VERTICAL SPACER, ITEM 4, WITH 2-10d NAILS.
5. NAIL LONGITUDINAL BOARD, ITEM 1, TO VERTICAL BOARD, ITEM 2, WITH 2-10d NAILS.
6. SIDE NAIL LATERAL BOARD, ITEM 3, TO VERTICAL BOARD, ITEM 2, WITH 2-10d NAILS.
7. NAIL UPPER BOARD, ITEM 6, TO LONGITUDINAL BOARD, ITEM 1, WITH 4-10d NAILS.



REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
4	6	UPPER BOARD	WOOD	2 X 4 X 27
4	5	HORIZONTAL SPACER	WOOD	2 X 4 X 17
4	4	VERTICAL SPACER	WOOD	2 X 4 X 19
4	3	LATERAL BOARD	WOOD	2 X 4 X 17
4	2	VERTICAL BOARD	WOOD	2 X 4 X 22
4	1	LONGITUDINAL BOARD	WOOD	2 X 4 X 115

LIST OF MATERIALS

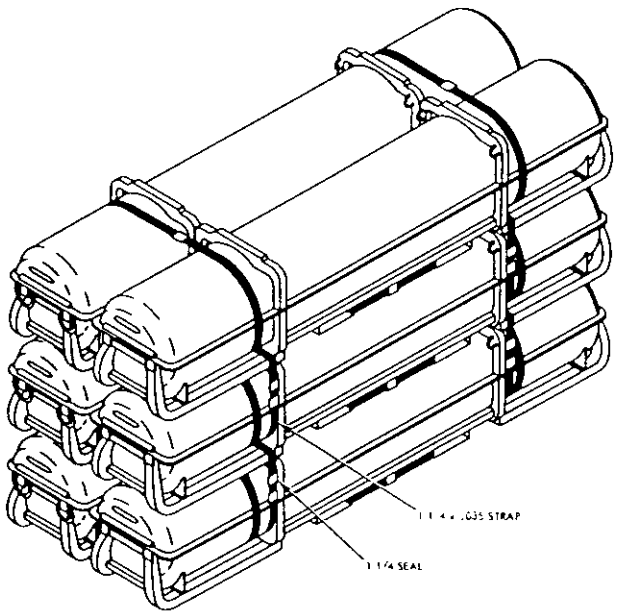
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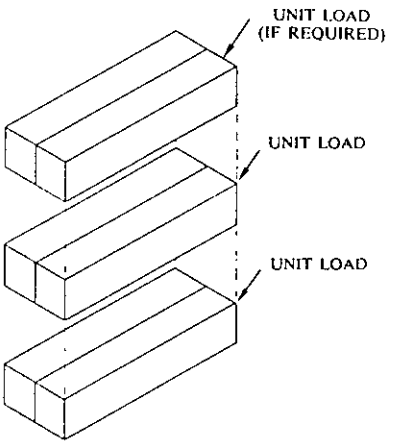
UNITIZING PROCEDURE

1. USING A FORK LIFT TRUCK, PLACE ONE UNIT LOAD ON TOP OF ANOTHER. INSURE THAT THE STACKING FEATURES ARE ENGAGED.

2. STRAP UNIT LOADS TOGETHER BY THREADING THE STRAPS, ITEM 1, UNDER THE ROUNDED SHELLS OF THE BOTTOM CONTAINERS AND THE SKIDS OF THE UPPER CONTAINERS. TENSION AND DOUBLE NOTCH SEAL USING ITEM 2.

3. REPEAT PROCEDURE FOR 3RD UNIT LOAD IF REQUIRED.





WARNING

UNITIZING UNIT LOADS IS RESTRICTED TO LOADING AND UNLOADING ONLY.

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
4	2	SEAL	STEEL	1 1/4 X .035
4	1	STRAP	STEEL	1 1/4 X .035 X 11 FT

LIST OF MATERIALS

REVIEW ACTIVITY:
NAVY-OS.

PREPARING ACTIVITY:
NAVY-OS
(PROJECT NO. 8140-N982)

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